Total No. of Questions : 4]

SEAT No. :

PD-266

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B.E. (Computer Engineering) (Insem)

DEEP LEARNING

(2019 Pattern) (Semester - VIII) (410251)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Answer Q1 or Q2, Q3 or Q4.***
2) ***Figures to the right indicate full marks.***
3) ***Neat diagrams must be drawn wherever necessary.***
4) ***Assume suitable data, if necessary.***

Q1) a) How Deep Learning works in three figures explain with example? Also explain common architectural principles of Deep Network? **[5]**

b) Short Note on LSTM networks & GRU networks **[5]**

c) Write a Short note on Autoencoders and Restricted Boltzmann Machines **[5]**

OR

Q2) a) Enlist and Explain any five popular industrial tools used for Deep Learning **[5]**

b) Explain Bias-Variance Tradeoff. **[5]**

c) Explain Convolution Neural Networks with Diagram. **[5]**

Q3) a) Enlist Activation functions used in Deep Neural Network. Explain any two of them in detail. **[5]**

b) What are the methods for tuning hyperparameters. **[5]**

c) What is the Loss function? Enlist all and Explain any two Loss function. **[5]**

OR

Q4) a) What is the Biological Neuron & Perceptron? What are the steps involved for training a Perceptron in Deep Learning? **[5]**

b) Explain how a Neural Networks can be trained with Back propagation and Forward propagation methods. **[5]**

c) Explain Gradient Descent. Why does the vanishing or exploding gradients problem happen? **[5]**